

SNN Off-Chip AER Interface Analysis

HW4AI Codefest 8

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Given

Output neurons: **N = 1024**

Mean firing rate: **f = 50 Hz**

Packet size: **20 bits** (10-bit address + 6-bit timestamp + 4-bit framing/parity)

Firing assumed independent across neurons (Poisson-like statistics).

Task 1: Mean Aggregate Spike Rate

$$R = N \times f$$

$$R = 1024 \times 50 \text{ Hz} = 51,200 \text{ spikes/s} \approx 51.2 \text{ kspikes/s}$$

Task 2: Mean AER Bandwidth

With 20 bits per packet (10 address + 6 timestamp + 4 framing/parity):

$$B = R \times 20$$

$$B = 51,200 \times 20 = 1,024,000 \text{ bits/s} \approx 1.024 \text{ Mbit/s}$$

Task 3: Interface Comparison (Mean Rate)

Interface	Capacity	Sustains 1.024 Mbit/s?
I ² C	≤ 3.4 Mbit/s	Yes (≈30% utilization)
SPI	≤ 50 Mbit/s	Yes (≈2%)
AXI4-Lite	~100 Mbit/s	Yes (≈1%)

Lowest-complexity interface that suffices: I²C. It clears the mean rate with comfortable headroom (~3.3× margin).

Task 4: Burst Analysis

A synchronous burst of 25% of neurons in a 1 ms window produces:

$$N_{burst} = 0.25 \times 1024 = 256 \text{ spikes in } 1 \text{ ms}$$

Peak instantaneous bandwidth:

$$B_{peak} = (256 \times 20 \text{ bits}) / 1 \text{ ms} = 5120 \text{ bits} / 10^{-3} \text{ s} = 5.12 \text{ Mbit/s}$$

Burst-to-mean ratio: $5.12 / 1.024 = 5\times$

Implication: I²C (3.4 Mbit/s) cannot absorb this burst in real time. It would back-pressure for roughly $5120 \text{ bits} / 3.4 \text{ Mbit/s} \approx 1.5 \text{ ms}$ to drain a 1 ms burst.

Two options:

- Step up to SPI (50 Mbit/s easily absorbs the burst, draining 5120 bits in $\sim 100 \mu\text{s}$).
- Keep I²C and add a FIFO buffer.

Minimum buffer depth $\approx$ burst size minus what drains during the burst:

$$Buffer \geq 256 - (3.4 \text{ Mbit/s} \times 1 \text{ ms} / 20 \text{ bits}) \approx 256 - 170 = 86 \text{ packets}$$

Round up to ~ 128 packets deep for safety margin against back-to-back bursts. Practically, SPI is the cleaner choice — it removes the buffering requirement entirely and still has $10\times$ headroom.

Task 5: Frame-Based Comparison

Frame readout: 1024 bits every 1 ms regardless of activity:

$$B_{frame} = 1024 \text{ bits} / 10^{-3} \text{ s} = 1.024 \text{ Mbit/s}$$

AER-to-frame ratio at mean rate:

$$B_{AER} / B_{frame} = 1.024 / 1.024 = 1.0$$

At $f = 50 \text{ Hz}$ with a 20-bit packet, AER and frame-based readout are essentially tied.

Crossover firing rate. Setting $B_{AER} = B_{frame}$:

$$N \times f \times 20 = N \times 1000 \text{ (samples/s)}$$

$$f_{crossover} = 1000 / 20 = 50 \text{ Hz}$$

Implication: AER wins when the mean firing rate is below $\sim 50 \text{ Hz}$ (sparse activity) — exactly the regime where most biologically-inspired SNNs operate. Above that, frame-based readout is more bandwidth-efficient because the per-spike framing overhead exceeds the cost of just sampling every neuron.